

Faithful Differentiated Expression in the Hilbert Substrate Framework: Expression-Term Policy for the Mesoscape Program

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Abstract

This note defines the policy for the expression term

$$\Delta E_{\text{expr}}$$

in the frozen lawful move law of the Hilbert Substrate Framework (HSF) mesoscape program. The immediate motivation is a concrete failure mode: once bookkeeping and birth-justification were improved, the model still exhibited monotone support growth to capacity, because the current expression term behaved too much like an additive reward for more edges or more support. This note replaces that interpretation with a stricter one.

The central claim is:

Expression gain is not raw summed relational weight. It is gain in *faithful differentiated local expression* of implicit relational structure in the constrained wavefunction, under a finite channel budget.

This yields four commitments:

1. Faithful expression means both that more implicit relational structure is made explicit locally and that the resulting support remains functionally distinguishable rather than redundant.
2. Channel novelty is primary; parent relief and sibling distinguishability are secondary but real.
3. Expression gain may go negative when added support becomes redundant and dilutes differentiated expression.
4. Removing redundancy can produce positive ΔE_{expr} , so lawful shell weakening and eventual retirement can compete without changing the frozen move law.

This note therefore defines the next proper target for bookkeeping and scoring: not raw support accumulation, but faithful differentiated expression under finite bandwidth.

1 Purpose

The lawful move law is already frozen:

$$\Delta F(M) = \Delta E_{\text{expr}} - \lambda_B \Delta C_B - \lambda_S \Delta C_S - \lambda_F \Delta C_F - \lambda_R W_{\text{NR}}. \quad (1)$$

So the present problem is not to invent a new event law. The problem is to define the term

$$\Delta E_{\text{expr}}$$

correctly.

Recent runs have shown the following pattern:

1. birth selection can be improved so that topological growth fronts are more plausible,
2. shell weakening can be classified lawfully,
3. retirement readiness can be made meaningful,
4. but the system still tends to fill the box and freeze.

The natural diagnosis is that the current expression term is still too additive: more support, more edges, more apparent expression.

This note replaces that with a different standard.

2 Guiding principle

The correct guiding principle is:

Guiding principle. ΔE_{expr} should measure gain in faithful differentiated local expression of implicit relational structure in the constrained wavefunction, not gain in raw support quantity or raw summed relational weight.

This implies three immediate exclusions:

1. Expression gain is not the same thing as more active nodes.
2. Expression gain is not the same thing as more active edges.
3. Expression gain is not the same thing as a larger sum of repeated copies of the same relational content.

3 Faithful expression

Faithful expression has two layers.

3.1 First-order faithful expression

The first-order question is:

Is previously implicit relational structure in the wavefunction being made explicit through local support that was not already representing it?

This is the primary sense in which birth can be good. A birth is justified when it increases the amount of implicit relational structure that is explicitly carried by the local support pattern.

3.2 Second-order faithful expression

The second-order question is:

Is the new support functionally distinguishable from existing supports already covering the same relational channel?

This is where mere replication fails.

If a new support patch simply duplicates a role already played by an existing sibling or nearby support patch, then it is not making new relational structure explicit. It is only re-expressing already expressed structure. In that case, raw relational weight may increase, but faithful differentiated expression does not.

Thus faithful expression must be both:

1. **expansive**: more implicit relational structure becomes explicit,
2. **differentiated**: the new explicit support plays a distinct local role.

4 Finite channel budget

The deepest saturation mechanism is finite bandwidth.

A parent pair or local interface sector has only a finite number of independent relational channels available for explicit support. For qutrit-level sectors, a natural upper bound is of order $d^2 = 9$ independent channel directions, though the exact physically active subset may be smaller.

This matters because once the existing children and neighboring support have collectively expressed the available channel structure between the parents, additional children cannot add genuinely new relational content. They can only:

- copy already-expressed roles,
- dilute distinctness,
- or increase raw support occupancy without increasing faithful differentiated expression.

So the primary saturation mechanism is:

Channel novelty saturates first.

Once channel novelty is exhausted:

1. parent relief saturates as a consequence,
2. sibling distinguishability saturates as a consequence.

Thus the root constraint is not merely “too many children,” but:

“the local finite channel budget is already fully represented.”

5 Three expression components

The expression term should therefore be decomposed into three conceptual components:

$$E_{\text{expr}}(R) = E_{\text{novel}}(R) + E_{\text{relief}}(R) + E_{\text{dist}}(R), \quad (2)$$

where:

- E_{novel} : **channel novelty** — how much previously unexpressed relational structure is now explicitly represented,
- E_{relief} : **parent / organizer relief** — how much constrained local stress is reduced by redistributing support,
- E_{dist} : **differentiated distinctness** — how distinguishable the local support roles remain after the move.

The policy adopted here is:

Priority ordering. Channel novelty is primary. Parent relief and support distinguishability are secondary but real.

So in any operational implementation, the weighting should satisfy

$$w_{\text{novel}} \geq w_{\text{relief}}, \quad w_{\text{novel}} \geq w_{\text{dist}}, \quad (3)$$

with all weights nonnegative.

6 Differentiated expression, not raw weight

The simplest wrong expression term is one that effectively computes:

more links carrying more correlator or MI weight $\Rightarrow$ more expression.

That is wrong because it ignores whether the added support is expressing new structure or merely duplicating existing structure.

The right object is closer to:

gain in the *dimensionality of explicit relational representation*, not gain in total relational weight.

In other words, the expression term should reward:

- new local channel directions made explicit,
- improved local role differentiation,
- improved faithful expression of parent or organizer structure,

not:

- duplicated siblings,
- repeated edge copies,
- support accumulation that leaves the organizer functionally unchanged.

7 Redundancy can make expression negative

A crucial consequence is:

Expression gain may be negative.

This is not an error. It is required.

If support becomes redundant, then it does not merely fail to add information. It can actively reduce the quality of local differentiated expression.

For example:

- suppose one support patch is uniquely expressing a local channel role,
- then a second support patch arrives as a near-identical copy,
- the region now contains two less-distinct supports instead of one clearly differentiated one.

In that case:

- total support may increase,
- total raw edge weight may increase,
- but differentiated local expression has decreased.

Thus redundancy can make

$$\Delta E_{\text{expr}} < 0.$$

This is the key mechanism that allows:

- redundant births to lose,
- shell weakening that removes duplication to gain positive expression credit,
- retirement of functionally evacuated support eventually to compete.

So flat saturation is not enough. If expression merely goes flat at redundancy, growth can stop but lawful compression still cannot win. Negative expression contribution from redundancy is therefore necessary.

8 Positive expression credit for removing redundancy

This gives the dual statement:

Removing redundant support can produce positive expression gain.

That includes two important cases.

8.1 Lawful shell weakening

Suppose a shell edge or shell support patch carries a relational role that is already fully represented through neighboring support. Weakening or reducing that redundant support can improve:

- distinctness of the surviving support,
- clarity of the local organizer role structure,
- faithful differentiated expression.

So lawful shell weakening may receive positive ΔE_{expr} , not merely lower burden.

8.2 Retirement after functional evacuation

If a support patch has already been evacuated of its distinctive organizer role, then retirement can improve differentiated expression by removing a redundant placeholder. That gives a route for retirement to become favorable under the existing frozen law, without inventing a special retirement bonus.

9 Birth justification in expression language

For birth moves, the expression term should incorporate a *birth justification witness*. The question is not:

Does the parent pair still allow more children?

but:

Does this new child add genuinely new differentiated local expression?

That witness should ask at least:

1. **Novelty:** does the proposed child express a channel role not already represented by nearby support?
2. **Parent relief:** does the child reduce overload or unresolved relational stress in the parent/organizer neighborhood?
3. **Sibling distinguishability:** is the child functionally different from existing siblings already attached to the same parent relation?

When the answer to these is largely no, the birth should lose expression credit.

10 Formal expression policy

The move-level expression difference should be treated as

$$\Delta E_{\text{expr}}(M; R) = \Delta E_{\text{novel}}(M; R) + \Delta E_{\text{relief}}(M; R) + \Delta E_{\text{dist}}(M; R), \quad (4)$$

with the policy that:

1. ΔE_{novel} is primary,
2. ΔE_{relief} and ΔE_{dist} are secondary but real,
3. each component may be positive or negative,
4. redundancy may drive $\Delta E_{\text{dist}} < 0$,
5. shell redundancy removal may drive $\Delta E_{\text{dist}} > 0$.

In a provisional normalized implementation one may write

$$\Delta E_{\text{expr}} = \eta_N \Delta N_{\text{expr}} + \eta_R \Delta R_{\text{expr}} + \eta_D \Delta D_{\text{expr}}, \quad \eta_N \geq \eta_R, \eta_N \geq \eta_D, \quad (5)$$

where:

- ΔN_{expr} is novelty gain/loss,
- ΔR_{expr} is parent or organizer relief gain/loss,
- ΔD_{expr} is differentiated distinctness gain/loss.

11 What does not belong in the expression term

The following do *not* count as faithful differentiated expression by themselves:

- raw node count,
- raw edge count,
- total support mass,
- repeated copies of the same sibling role,
- larger summed relational weight with no added channel novelty or role distinctness.

These may correlate with expression in some regimes, but they are not the definition.

12 Relation to burden terms

This note does not move all compression into the expression term.

The burden terms remain necessary:

- ΔC_B should still encode finite bandwidth pressure,
- ΔC_S should still encode spread pressure,
- ΔC_F and W_{NR} should still guard lawful identity and no-refolding.

But the expression term must stop rewarding support merely for existing. Otherwise the burden terms are forced to do impossible work against a monotone additive reward.

So the correct division of labor is:

- expression rewards faithful differentiated local representation,
- bandwidth and spread penalize redundant or over-broad support load,
- no-forgetting and no-refolding determine whether compression is lawful.

13 Failure conditions

This policy fails if one or more of the following occur:

1. ΔE_{expr} is still effectively monotone in support count or edge count.
2. Redundant sibling births keep receiving positive expression credit after channel novelty is exhausted.
3. Shell weakening that removes clearly redundant support cannot ever produce positive expression credit.
4. Retirement of functionally evacuated support can never improve expression.
5. The expression term collapses to raw summed MI / correlator / link activity with no distinguishability correction.

If these occur, then the expression term has not been successfully rewritten.

14 Immediate implementation consequences

This note implies the following implementation discipline.

1. Birth scoring must include channel novelty, parent relief, and sibling distinguishability.
2. Repeated children off one parent pair must receive diminishing and eventually negative expression credit once available channel novelty is exhausted.
3. Shell weakening should be allowed to receive positive expression credit when it removes redundant support while preserving organizer function.
4. Retirement should remain gated by organizer evacuation, but once redundancy is high enough it may gain positive expression credit from removing functionally duplicate support.
5. Raw additive local-expression sums must be replaced or corrected by a differentiated-expression measure.

15 Conclusion

The mesoscape program should not treat expression as the quantity of support or the total weight of repeated relational representation. It should treat expression as faithful differentiated local realization of implicit relational structure under a finite channel budget.

That means:

- channel novelty is primary,
- parent relief and sibling distinguishability are real,
- redundancy can make expression negative,
- and removing redundancy can improve expression.

This is the right policy for the next rewrite of ΔE_{expr} . It preserves the frozen lawful move law while finally giving the model a route from mere support growth toward lawful compression and eventual organizer retirement.